

Memory-Channel and Mode-Locking Constraints in TRM/TQM V3.0

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Abstract

This paper isolates two tightly coupled TRM/TQM V3.0 tracks: the memory-channel derivation path and the rational mode-locking closure path [7, 5]. The goal is to document what is currently tested-effective and structurally constrained, while preserving explicit claim boundaries [2]. The memory term $\phi^2|\dot{\mu}|$ is currently strongly supported as a minimal effective invariant under guard tests, and the $m = 3$ closure route is currently strongly constrained through RBF hardening [6, 8]. Neither path is claimed as theorem-level microscopic closure.

1 Introduction

TRM/TQM V3.0 uses a test-gated strategy: derive candidate structures, stress them under ablations, and retain only forms that survive structural and numerical constraints [2]. Within this strategy, the memory-channel and mode-locking blocks are central because they constrain transport response and closure order, respectively.

This document focuses on:

1. memory-channel admissibility and current microscopic support level;
2. mode-locking closure structure and the current $m = 3$ theorem-path hardening;
3. explicit failure criteria and remaining first-principles gaps.

2 Memory-Channel Track

2.1 Core object

The effective memory-channel invariant is currently represented by:

$$\phi^2|\dot{\mu}|.$$

2.2 Current test support

The memory path is guarded by MEM/TRM/MC blocks, including MC09-MC12 hardening, and currently supports [7, 6, 4, 3]:

$$A_{\text{dyn}} \propto \phi \rightarrow A_{\text{dyn}}^2|\dot{\mu}| \rightarrow \phi^2|\dot{\mu}|.$$

At V3.0, this is interpreted as strong lattice-proxy/microscopic support at effective level, not theorem-level derivation.

2.3 Claim boundary

- Supported: tested-effective/hypothesis-supported effective invariant path.
- Not claimed: theorem-level microscopic uniqueness proof.

3 Rational Mode-Locking and $m = 3$ Closure Track

3.1 Core object

The mode-locking path is organized around closure-family constraints and the inverse rational band [1, 8]:

$$\Omega = \frac{q+3}{q}, \quad \gamma = \frac{1}{\Omega}$$
$$\Omega \approx 1.16..1.19, \quad \gamma \approx 0.84..0.86.$$

3.2 Current hardening status

The Gap-2 path is currently hardened through RBF blocks, with RBF16-RBF23 providing [5, 2]:

1. connected threshold-region stability,
2. explicit failure-by-family exclusion,
3. bounded perturbation stability,
4. microscopic phase-lattice-energy-based action/tick discriminator reinforcement.

This keeps $m = 3$ as a strongly constrained closure-order candidate under the tested rule family.

3.3 Claim boundary

- Supported: strongly constrained theorem path for $m = 3$.
- Not claimed: theorem-level microscopic closure theorem.

4 Coupled Interpretation: Memory + Mode-Locking

The two tracks are methodologically coupled:

1. the memory-channel block constrains admissible transport-memory structure;
2. the mode-locking block constrains admissible closure order and discriminator structure.

Together they reduce arbitrary model freedom, but they do not yet establish full microscopic first-principles completion.

5 Falsification and Failure Criteria

The current interpretation should be narrowed or rejected if:

1. a competing memory invariant satisfies the same admissibility constraints with strictly better bridge behavior and no additional penalties;
2. $m = 3$ loses constrained minimality under robust, non-ad-hoc perturbation families;
3. microscopic derivation assumptions force a different leading invariant or closure order under the same structural constraints.

6 Remaining Gaps

Open items remain explicit:

- theorem-level microscopic closure for $\phi^2|\dot{\mu}|$,
- theorem-level microscopic closure for $m = 3$,
- full first-principles derivation bridge between these blocks and broader unified action structure.

7 Conclusion

In V3.0, the memory-channel and mode-locking tracks are substantially hardened and claim-safe: both are strongly constrained and tested-effective in their current role, but neither is promoted to theorem-level microscopic proof. This status supports critical external review and targeted next-step falsification work, not over-claiming completion.

References

- [1] TRM Project. Trm collective mode locking bridge scale, 2026. Repository document: docs/Theory/TRM_Collective_Mode_Locking_BridgeScale.md.
- [2] TRM Project. Trm current status for peer review, 2026. Repository document: docs/review/TRM_Current_Status_For_PeerReview.md.
- [3] TRM Project. Trm finsler optical action, 2026. Repository document: docs/Theory/TRM_Finsler_Optical_Action.md.
- [4] TRM Project. Trm geodesic derivation, 2026. Repository document: docs/Theory/TRM_Geodesic_Derivation.md.
- [5] TRM Project. Trm m3 closure theorem path, 2026. Repository document: docs/Theory/TRM_M3_Closure_Theorem_Path.md.
- [6] TRM Project. Trm memory channel: First-principles derivation track, 2026. Repository document: docs/Theory/TRM_Memory_Channel_First_Principles.md.
- [7] TRM Project. Trm memory channel microscopic derivation, 2026. Repository document: docs/Theory/TRM_Memory_Channel_Microscopic_Derivation.md.
- [8] TRM Project. Trm rational band: First-principles origin, 2026. Repository document: docs/Theory/TRM_Rational_Band_First_Principles.md.